# CryptoGuesser

AI-powered cryptocurrency price direction predictor · 3-model LSTM ensemble trained on Binance market data

April 2026
**Python 3.12** · PyTorch · Streamlit
CUDA 12 (RTX recommended)

### WHAT IT DOES

Predicts whether a token's next candle will close **UP** or **DOWN** using three LSTM models at different timeframes, combined into an ensemble signal. Only models above the **65% confidence threshold** vote.

### SYMBOLS SUPPORTED

BTC, ETH, BNB, SOL, ADA, AVAX, DOT, LINK, XRP, MATIC — all paired against **USDT**, fetched directly from **Binance** via CCXT.

### THREE MODELS

**M1** — 1h candles, 6 months. **M2** — 4h candles, 1 year. **M3** — 1d candles, 3 years. Each captures a different market pattern horizon.

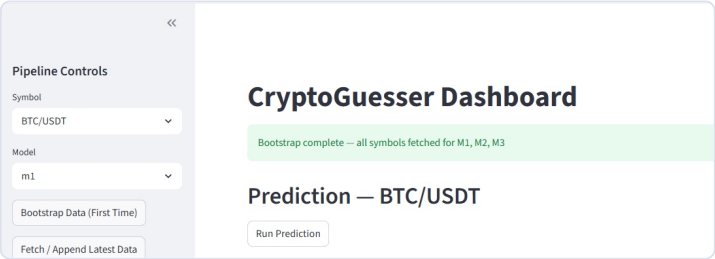
| Model | Candle size | History | Lookback window | What it captures |
|---|---|---|---|---|
| **M1** | 1 hour | 6 months | 96 candles | Short momentum, hourly patterns |
| **M2** | 4 hours | 1 year | 84 candles | Weekly cycles, trend continuation |
| **M3** | 1 day | 3 years | 60 candles | Macro trend, bull/bear phases |

### ① DATA GATHERING

A one-time bootstrap downloads the full candle history for all 10 symbols across M1, M2, and M3. Data is stored as local **Parquet files** per symbol per model. The daily append command fetches only new candles and merges them in, keeping the dataset current without full re-downloads.

The dashboard exposes both actions directly — Bootstrap is used on first setup, Fetch/Append for daily updates.

ccxt / Binance · Parquet storage · M1 · M2 · M3




### ② MODEL TRAINING

Each model uses **three stacked LSTM layers**, multi-head self-attention, layer normalisation, and two dense layers ending in a binary sigmoid output. Features include RSI, MACD, Bollinger Bands, ATR, Stochastic, EMA crossovers, OBV, and log returns — **~20 features** in total, scaled with a saved StandardScaler.

Training uses Adam, ReduceLROnPlateau, gradient clipping, class-balanced BCE loss, and early stopping (patience = 20). A Val AUC > 0.65 is considered a strong signal for crypto prediction.

Val AUC displayed · Early stopping · Class balancing

### ③ PRICE PREDICTION

Selecting a symbol and model in the sidebar then clicking Run Prediction loads the latest checkpoint, applies the saved scaler to the most recent candle window, and returns a directional signal with a confidence score.

If confidence is below the 65% threshold, the signal is flagged as unreliable. The ensemble combines all three models — if none are confident, the system **abstains** rather than guessing.

UP / DOWN signal · Confidence score · Ensemble voting · Abstain if unsure